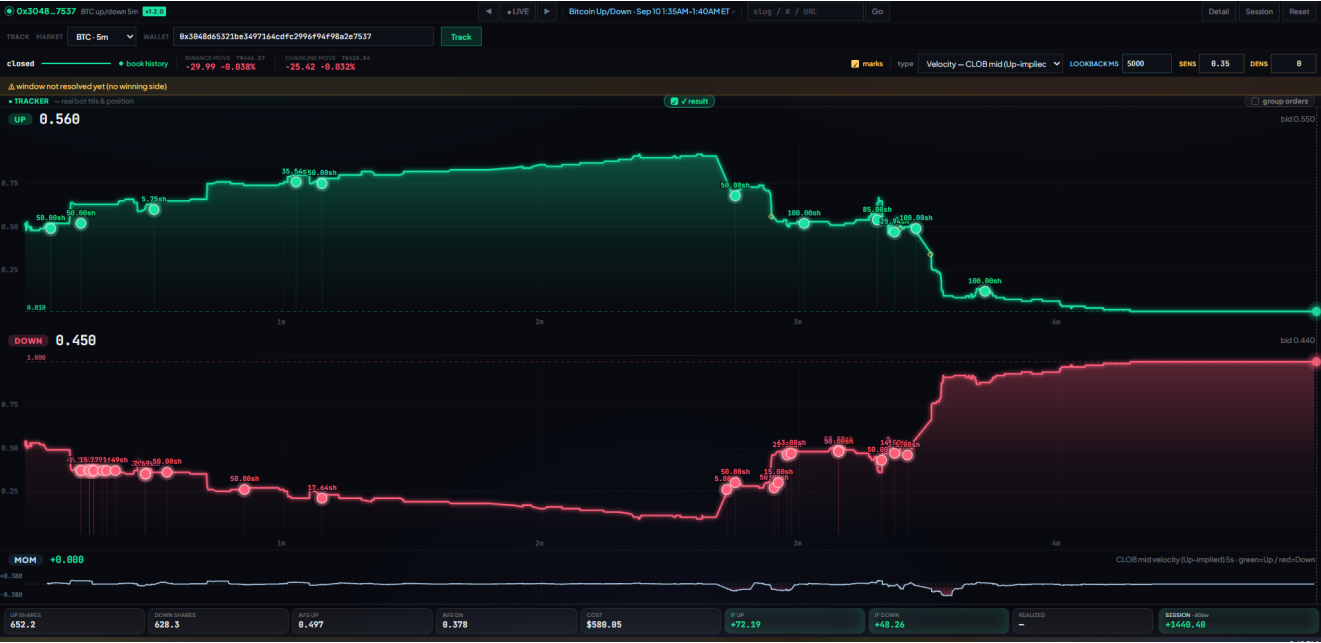


Adaptive Inventory and Target-Price Sizing

Order quantities are derived from target settlement payoffs, acquisition averages, signed inventory imbalance and attainable prices. There is no preset per-round capital ceiling or bankroll-scaled Kelly term in this formulation. Expenditure is an output of the selected quantities. Favorable prices can support additional accumulation without reference to an arbitrary dollar allocation.



The supplied image illustrates an inventory whose displayed conditional P&L is positive under either settlement. It does not establish that the round has resolved; the dashboard explicitly says the winner is unresolved. The session total is separate from this round.

Visible metric	Value
UP / DOWN shares	652.2 / 628.3
Average UP / DOWN token price	0.497 / 0.378
Displayed total cost	\$580.05
Displayed IF UP / IF DOWN	+\$72.19 / +\$48.26
Share DIFF / ratio from rounded quantities	23.9 shares / 1.0380

Calculations using the rounded holdings and displayed cost give +\$72.15 and +\$48.25. The small differences from the dashboard can be explained by undisplayed precision; the unrounded ledger is needed to reconcile them. The example is evidence for the importance of total acquisition cost relative to both quantities, not evidence that a 1.2-1.3 holdings ratio is necessary.

This report replaces the earlier capital-based framing with target-payoff equations, price-dependent quantity formulas and staged execution. Its exact algebra is verified numerically. Future target prices and real maker fills remain uncertain; no market-optimal next order is inferred from a completed chart alone.

Profitable settlement in both directions is an inventory-cost condition.

Symbol	Definition
U, D	Confirmed UP and DOWN quantities; use unrounded net credited shares.
C	Total acquisition cost including applicable execution fees and other costs included in the round ledger.
$C_U^{\text{token}}, C_D^{\text{token}}$	Token purchase expenditures by side, before separately recorded fees.
A_U, A_D	Token-price averages C_s^{token}/N_s . Track fee-inclusive averages separately as well.
P_U, P_D	Conditional settlement P&Ls $U - C$ and $D - C$.
$\Delta; H$	Signed difference $U - D$; current common settlement payoff $\min(U, D) - C$.
G_U, G_D	Desired conditional P&Ls at a selected future inventory point. These are planning targets, not booked profits.

$$P_U = U - C, \quad P_D = D - C, \quad H = \min(U, D) - C$$

$$H > 0 \iff C < \min(U, D)$$

Using the rounded image values, $H = \$48.25$ and $\Delta = 23.9$. UP pays \$23.9 more than DOWN because the position contains that many additional UP shares. The positive common payout comes from acquiring the total inventory for less than either settlement quantity.

The displayed token averages imply about \$561.64 of raw acquisition cost, versus \$580.05 displayed total cost. The \$18.41 difference cannot be assigned to fees or other adjustments from the image alone. Use the actual cost ledger for P&L; do not infer net profitability from the two displayed averages alone.

When an exact target position is reached, $U' = C' + G_U$ and $D' = C' + G_D$. Consequently:

$$\Delta' = G_U - G_D, \quad \frac{U'}{D'} = \frac{C' + G_U}{C' + G_D}$$

The holdings ratio can therefore be an output of the intended payoff structure. Giving the currently favored outcome the larger payoff target gives it more shares automatically. Large common paired inventory can support a ratio close to one while both settlement profits remain meaningful.

A fixed initial-side label and a current winning-side prediction are different objects. Preserve the initial label for evaluation. If current predictions are permitted to change, switch the favored target assignment without deleting or resetting any existing inventory or costs. If initial-side dominance is retained as a policy, a later forecast reversal does not by itself reverse holdings.

A target price translates into quantities through explicit payoff and average equations.

Let $K_U(x)$ be the all-in cost of an UP purchase of x shares. Its post-fill payoffs are:

$$P'_U = P_U + x - K_U(x), \qquad P'_D = P_D - K_U(x)$$

For a DOWN purchase of y shares, $P'_U = P_U - K_D(y)$ and $P'_D = P_D + y - K_D(y)$. A single-side buy always transfers payoff capacity from the other settlement outcome to its own. The desired quantity must reflect that transfer.

At a constant all-in UP acquisition cost a , a desired UP payoff G_U requires:

$$x_{\text{required}} = \max\big(0, \frac{G_U - P_U}{1 - a}\big)$$

If a selected DOWN payoff floor G_D is also to be preserved, x must satisfy $ax \leq P_D - G_D$. This is a consequence of the requested payoff structure, not a capital ceiling. If the lower and upper quantities conflict, that one-sided trade cannot achieve both targets. A different price, paired plan or revised target is necessary.

For a desired lower token-price average A^* on a side holding N shares at average A , a maker fill at token price u gives the exact quantity:

$$q_{\text{average}} = \frac{N(A - A^*)}{A^* - u}, \qquad u < A^* < A$$

UP target average from 0.497 to 0.480	Required additional UP shares
Maker price \$0.20	39.60
Maker price \$0.30	61.60
Maker price \$0.40	138.59

The table uses the image’s 652.2 UP shares. Cheaper tokens require **fewer** shares to reach the same average target. If the intent is to buy more at cheaper prices, that corresponds to choosing a more ambitious payoff or average target, not an algebraic requirement that every lower price must trigger a larger order.

If $u \geq A$, no positive purchase quantity can lower the raw token-price average. For fee-inclusive averages, replace u by all-in cost and A by the corresponding fee-inclusive average. A rising-price taker purchase can still reduce the average if it occurs below that average; increasing size cannot solve an above-average acquisition price.

For multiple levels, keep each level’s cost in $K(q)$. If every intermediate fill must preserve the average, every independently possible fill must pass the average test. Do not rely on a future cheap fill to offset an expensive fill that has already arrived.

Solve both quantities together when the target includes both outcomes.

For proposed all-in acquisition prices a for UP and b for DOWN, let x and y be the additional quantities. Define $d_U = G_U - P_U$ and $d_D = G_D - P_D$. The exact target equations are:

$$(1 - a)x - by = d_U, \quad -ax + (1 - b)y = d_D$$

$$x = \frac{(1 - b)d_U + bd_D}{1 - a - b}, \quad y = \frac{ad_U + (1 - a)d_D}{1 - a - b}$$

Check nonnegative quantities, exchange precision, available liquidity and all intermediate fills. Negative quantities indicate that the exact target cannot be reached with these BUY-only legs; solve payoff inequalities instead if overachieving a target is acceptable.

For positive improvements in both settlement outcomes, the constant-price two-leg plan needs $a + b < 1$. At $a + b = 1$ there is no positive combined margin; above one, scaling both legs cannot improve both outcomes. These are fee-inclusive prices. In a sequential strategy the two prices may occur at different times, so their joint availability must be modeled rather than assumed.

From the image, choose illustrative targets $G_U = \$100$ and $G_D = \$70$. These amounts demonstrate the calculation; they are not proposed fixed production settings. Two candidate price combinations produce:

Candidate future fills	Additional UP	Additional DOWN	Additional cost
\$0.30 UP / \$0.20 DOWN	53.26	47.16	\$25.41
\$0.49 UP / \$0.37 DOWN	182.81	176.71	\$154.96

Both combinations are below the respective displayed averages, and both reach the same intended $+\$100/+\70 payoffs in the constant-price full-fill model. Cheaper acquisition accomplishes the target with much less additional inventory. Extra favorable depth could instead be used to raise the payoff target; there is no preset per-round expenditure cap preventing that choice.

For the first candidate, the final inventory is 705.46 UP and 675.46 DOWN; total cost is \$605.46. Raw UP average falls to about 0.48213 and DOWN average to 0.36557. The new DIFF is 30, precisely the difference between the two payoff targets.

These are hypothetical future fill prices, not available quotes verified from the screenshot. The chart alone cannot establish that either two-leg plan could actually have been executed, or that its prices would be attainable again before settlement.

The correct quantities still require an executable fill sequence.

Submitting the whole planned UP leg first would temporarily reduce the DOWN payoff to about \$32.27. Submitting the whole DOWN leg first would temporarily put more shares on DOWN than UP. Neither behavior is visible in the final +\$100/+\$70 target alone.

A staged sequence can preserve UP dominance and the image's existing \$48.25 common payoff throughout, provided the hypothetical target fills become available in this order:

Confirmed fill stage	UP shares	DOWN shares	IF UP	IF DOWN
Initial	652.20	628.30	\$72.150	\$48.250
DOWN 20 @ 0.20	652.20	648.30	\$68.150	\$64.250
UP 53.26 @ 0.30	705.46	648.30	\$105.432	\$48.272
DOWN 27.16 @ 0.20	705.46	675.46	\$100.000	\$70.000

First, 20 DOWN shares at \$0.20 create \$16 of additional DOWN-payoff room. Then 53.26 UP shares at \$0.30 consume \$15.978 of that room. Finally, the remaining 27.16 DOWN shares at \$0.20 complete the target. Both averages fall by the end; all intermediate positions retain more UP shares and remain above the initial common payoff.

The first DOWN quantity is itself calculable. To preserve a common payoff floor H during the later UP purchase x , a preliminary DOWN fill z at cost b must satisfy:

$$z \geq \max\left(0, \frac{ax - (P_D - H)}{1 - b}\right)$$

With $H = P_D = \$48.25$, $a = 0.30$ and $x = 53.26$, $z \geq 19.9725$. Rounding upward to 20 makes the illustrated sequence feasible, while the existing UP advantage of 23.9 shares permits that initial DOWN quantity. Actual venue quantity precision and minimum sizes must be checked before rounding.

This floor is an illustrative choice to preserve an already achieved profit, not a newly imposed capital budget. If the objective permits surrendering some current common profit to improve a forecasted return, show the trade-off explicitly rather than silently assuming the future leg will restore it.

Recompute after every actual partial fill. Use only confirmed quantities and costs. An accepted order, a touched price or a predicted fill is not confirmed inventory. Pending quantities must remain reserved against the adverse fill sequence until matched, canceled or otherwise reconciled.

The same calculations apply when DOWN is favored by swapping the roles of the outcomes. If prediction changes during the sequence, recompute target assignments and residual orders; do not continue a stale plan merely because its first leg filled. The example verifies safe intermediate arithmetic, not future price availability.

Prediction points should describe joint price paths and fill opportunities.

The decision state should contain confirmed U/D quantities; cost and side averages; signed DIFF; both conditional P&Ls; pending orders; current bid/ask ladders; the market's reference-price gap; token-price velocities; gap velocity; and remaining time. Keep exchange observations, receipt times and decision times separate.

$$g_t = R_t - R_{\text{open}}, \quad v_t^g = \frac{g_t - g_{t-h}}{h}, \quad v_t^s = \frac{m_t^s - m_{t-h}^s}{h}$$

R is the exact settlement-reference stream specified for the market, and m^s is the outcome midpoint or another explicitly defined token-price measure. Actual execution costs come from the book rather than midpoint forecasts. Polymarket publishes Chainlink TWAP feeds with separate observation timestamps; the complete custom TWAP calculation details are not publicly specified.¹

For each usable lookahead h before the round ends, estimate a conditional distribution of executable prices and fills from the current market state. Candidate price points may be selected from that distribution or from actual quote levels. Each candidate pair of UP/DOWN price points produces candidate quantities through the target equations.

Pr(UP fill at a , DOWN fill at b , order of fills | x_t)

The joint distribution is essential. Treating the probabilities of a cheap UP fill and a cheap DOWN fill as independent can create fictitious paired opportunities. A path may reach one target while moving permanently away from the other, especially late in the round.

A simple projected price based on current velocity can generate candidate points, but it is not a validated probability model. Estimate forecast errors and target-hit frequencies on unseen rounds. Maker fill probability also depends on queue position and actual opposing trade flow; target price touch alone is insufficient.

For the retained maker rule, BESTASK minus one tick is the executable quote candidate at the moment of submission. A deeper predicted price is a future trigger point: under this fixed placement rule, wait until the current quote reaches an economically acceptable level. Submitting deeper standing quotes would be a separately evaluated placement policy.

Recalculate the plan when a fill, cancellation, meaningful book change, gap change, direction change or passage of time materially changes feasibility or expected value. Recalculation need not mean replacement of every standing order on every update; unnecessary replacement can sacrifice queue position.

At the round deadline, the outcome is determined under the market rules; official resolution and redemption can follow later. As remaining time shrinks, the probability of completing a sequential plan must be reassessed. Do not manufacture a final-window recovery by assuming an opposite fill beyond the deadline.

1. Polymarket, Chainlink TWAP Prices, accessed September 10, 2026.

Price limits follow from the target inventory and execution cost.

For proposed quantities x and y with total incremental cost $K = K_U(x) + K_D(y)$, the target-payoff frontier is:

$$K \leq \min(P_U + x - G_U, P_D + y - G_D)$$

At constant prices this becomes $ax + by \leq \min(P_U + x - G_U, P_D + y - G_D)$. For fixed y and b , it gives the maximum admissible UP cost; the symmetric expression gives the DOWN cost. Intersect this frontier with the average-price conditions and any required favored-side dominance. The resulting ceiling varies with inventory and target, rather than remaining at \$0.50 or \$0.89.

If a single opposite-side purchase is intended to make both outcomes nonnegative while $U > D$, assume UP is favored and the current UP payoff is positive. A DOWN buy at all-in cost b must satisfy:

$$\max\left(0, \frac{-P_D}{1-b}\right) \leq y \leq \frac{P_U}{b}$$

When $P_D < 0$, feasibility requires $b \leq P_U/(P_U - P_D) = P_U/\Delta$; strict profit requires appropriate strict margins. Preserving a positive favored share advantage additionally limits y to less than Δ . This is a repair-price frontier obtained from existing inventory, with no round-capital parameter.

For example, 100 UP shares with total cost \$60 give +\$40/- \$60. Buying 90 DOWN at \$0.20 raises total cost to \$78 and gives +\$22/+ \$12, with more UP shares. A DOWN purchase at \$0.50 cannot make both outcomes positive from that starting position. The difference is the feasible payoff geometry, not an arbitrary token-price cutoff.

At currently available taker levels, compute cost from actual depth and price protection. Published crypto fees are $0.07 \times \text{quantity} \times \text{price} \times (1 - \text{price})$, subject to rounding; makers are not charged trading fees.² Use the exact applicable market parameters. If the modeled cost includes fees, never add them a second time in settlement.

Maker orders should use post-only to prevent immediate taking. Taker intentions can use a bounded marketable limit and FAK when unfilled quantity should be canceled.³ The order type is selected by its predicted execution contribution to the same inventory plan, not by an independent size rule.

A rising favored price may justify taking when the price is still below the relevant average and waiting is expected to miss a useful fill. If it is above the average, no quantity can satisfy a strict average-reduction requirement. That conflict must lead to waiting or an explicit change in the objective, not a hidden exception.

2. Polymarket, Fees. 3. Polymarket, Place Orders, and Order Lifecycle.

The controller selects a reachable payoff frontier, then updates it as fills arrive.

Without a preset capital ceiling, a finite next quantity can still come from the inventory target, actual executable depth, the deadline and marginal economics. If favorable pair prices had unlimited depth forever, profit would scale without bound; no finite global maximum would exist. This is a mathematical property of unlimited positive-margin acquisition, not a reason to reintroduce a round spending cap.

Use a finite set of actual or statistically modeled execution opportunities. The controller should construct a frontier of attainable conditional payoffs and average costs. A plan dominates another when it improves both settlement payoffs with no worse required average-price behavior and no worse execution assumptions. Do not rank an uncertain two-leg projection as though it were already filled.

$$E[P_T | x_t] = E[U_T Y + D_T(1 - Y) - C_T | x_t]$$

The expectation must be taken over the joint distribution of settlement, changing prices, quantities and fills. Substituting $E[U] \times P(UP)$ for $E[UY]$ is generally invalid: the size of the eventual position can be correlated with the outcome. This is especially important for averaging strategies that accumulate their largest positions in adverse paths.

Stage	Required calculation
Reconcile	Update filled shares, side costs, averages and both P&Ls. Preserve pending-order reservations.
Forecast	Generate attainable price/fill paths from the gap, token movements, books and remaining time.
Construct	Calculate quantities for candidate payoff and average targets; include no order.
Sequence	Check partial fills and the order of legs, favored dominance and any selected preservation of current common profit.
Rank	Compare expected net P&L, conditional downside and attainable averages; reject dominated plans.
Submit	Place only the currently admissible next leg at its calculated quote and quantity.
Recalculate	Rebuild the remaining plan after actual changes instead of treating targets as fixed instructions.

The stated goal of maximizing average profit while minimizing losses is multi-objective. A point on the frontier still requires a preference when one action offers more expected return but worse downside. Prefer plans improving both outcomes where feasible. When none exists, report the trade-off rather than claiming there is one uniquely optimal size independent of probabilities and objective choice.

Once both outcomes are profitable, preserving some or all of the current common payoff is one possible preference. Before reaching that state, distinguish a projected completion target from the current exposed position. A future cheap fill can be planned but cannot be used as evidence that the current position is already protected.

Choose targets by marginal payoff value, rather than choosing quantities from price alone.

One explicit way to rank the return/downside frontier is the terminal score below, with a nonnegative, dimensionless loss weight λ . This is an optional preference parameter, not a capital limit. $\lambda = 0$ selects expected net profit; larger λ assigns extra cost to negative round outcomes. The data estimate outcomes; the desired trade-off selects λ .

$$\psi_\lambda(z) = z - \lambda \max(-z, 0), \quad J = E[\psi_\lambda(P_T) \mid x_t]$$

For certain immediate fills and a fixed calibrated UP probability p , $J = p \psi(P'_U) + (1-p) \psi(P'_D)$. Away from zero-payoff crossings, define the marginal weights:

$$w_U = p[1 + \lambda \mathbf{1}_{P_U < 0}], \quad w_D = (1-p)[1 + \lambda \mathbf{1}_{P_D < 0}]$$

If a and b are all-in marginal acquisition costs, the local score gain per additional share is:

$$\frac{dJ}{dx} = w_U - (w_U + w_D)a, \quad \frac{dJ}{dy} = w_D - (w_U + w_D)b$$

$$a < \frac{w_U}{w_U + w_D}, \quad b < \frac{w_D}{w_U + w_D}$$

These are local marginal-value ceilings before binding dominance, average-price or staged-payoff constraints. For $\lambda = 0$ they reduce to $a < p$ and $b < 1-p$. If the DOWN outcome is losing while UP is profitable, extra downside weight raises the admissible DOWN price because that purchase repairs a negative outcome. A \$0.50 ceiling is therefore not universal. Intersect the economic ceiling with the strict average-reduction condition, valid \$0.01-\$0.99 quote levels and actual depth.

Increase quantity through available price levels while the constrained marginal score is positive. Recompute at payoff-zero crossings, a change in marginal book price, a dominance boundary or a changed forecast. A cheap token whose probability falls even faster can have a negative score; an old average price does not establish current undervaluation. When a side has no holdings, its first purchase establishes its average rather than lowering an existing one.

For maker orders and multi-step plans, use the joint execution model instead of the certain-fill shortcut. The full decision is a dynamic program, with terminal value ψ of actual round P&L:

$$V_t(x) = \max_{o \in \mathcal{A}(x)} E[V_{t+\delta}(X_{t+\delta}) \mid x, o]$$

An action o includes side, price, size and maker/taker mode; waiting is admissible. The feasible action set enforces current average and favored-share requirements and actual exchange mechanics, with no preset spending cap. In a scenario approximation, future actions sharing the same observed history must be identical: they cannot use knowledge of which future path will occur. Solve on a rolling horizon and submit only its first action. Target payoffs and DIFF can then be outputs of the selected plan; the inverse equations explain and verify its quantities.

The five-minute probability model must be calibrated against the settlement reference and execution.

A useful mathematical baseline for the reference gap g is a diffusion with drift μ and volatility σ per square-root second. If its remaining change is Normal($\mu\tau$, $\sigma^2\tau$), the provisional UP probability is:

$$p_t \approx \Phi\left(\frac{g_t + \mu_t\tau}{\sigma_t\sqrt{\tau}}\right), \quad \tau = 300 - t$$

This is a modeling assumption, not an established description of BTC or the market's oracle. The exact opening/end reference, any averaging window and tie rule must match the specific contract. In volatile periods, jumps, changing volatility and stale measurements can invalidate the Gaussian shortcut. Fit the conditional distribution of the actual settlement-reference change, or calibrate a richer probability model, before using these values for sizing.

The normalization explains why the same dollar gap and token velocity can require different decisions at 20 versus 280 elapsed seconds. A raw gap is insufficient without remaining uncertainty. UP and DOWN movements are also strongly related by their complementary payoffs; treating both velocities as independent evidence can exaggerate confidence.

Estimate jointly: settlement probabilities; arrival-price/slippage distributions; maker fill quantity conditional on quote, queue and flow; and the probability of completing a sequence before expiry. A maker fill is an informative event, so the appropriate success probability after filling can differ from the probability at submission. The scalar initial-entry accuracy is not a substitute for any of these state-dependent quantities.

Attached v5 replay statistic	Reported result
First-entry correctness	453 / 860 = 52.67%
Profitable complete rounds	316 / 860 = 36.74%
Average positive / negative round	+\$51.11 / -\$65.83
Mean complete-round P&L	-\$22.86
Correct first entry but negative round	208 rounds

These are reported results from the supplied specification, not a new replay. Its main study disabled resting maker fills and used fixed size templates. It therefore supports the need to evaluate complete inventory construction, but it does not validate the proposed BESTASK-minus-one-tick maker policy. The newer quoted 51.48% is a separate statistic without a supplied denominator and must not be silently substituted into this sample.

$$E[P_T] = p_+ E[P_T | P_T > 0] - p_- E[-P_T | P_T < 0]$$

If all nonzero rounds are classified, profitability requires the average-win/average-loss ratio to exceed p_-/p_+ . A 51.48% *profitable-round* rate would imply a break-even ratio of about 0.9425; a 51.48% *entry-correctness* rate does not imply that threshold for the actual trading strategy.

Validate on chronological held-out rounds with fees, cancellations, partial fills and realistic latency. Report mean net P&L, loss frequency, average and tail losses, target-completion frequency and calibration by remaining time. Use block resampling over contiguous rounds or days for uncertainty; multiple ticks in one round share one outcome. Select model parameters inside the training/validation period and reserve an untouched test period. Optimize the return/downside frontier, not entry accuracy alone.

Verification, empirical limits and final specification

The two-leg inverse equations were checked over 2,000 random price/target cases with $a + b < 1$. In 500 of those cases, the acquisition cost matched an independently solved linear program for the same payoff inequalities; maximum cost difference was 2.73×10^{-12} . One thousand target-average calculations also reproduced their specified average. The illustrated three-stage sequence was checked for both positive settlement P&Ls and UP dominance at every stage.

Those checks establish the mathematical and numerical consistency of the equations. They do not establish real queue fills, future target prices, live profitability or optimal model coefficients. The screenshot contains a successful conditional end-position, not the data needed to reconstruct each order-placement decision.

Empirical validation needs native timestamped books and trades, full order and cancellation records, fees, unrounded inventory, point-in-time opening references and final settlement. Estimate target-hit and fill distributions by remaining time and market state. Validate chronological periods separately; do not treat many updates sharing one five-minute outcome as independent settlement samples.

The 51.48% initial-direction figure remains a historical cohort statistic with no accompanying sample count in the current material. Final profitable-round frequency can differ from it. The realized profitability identity remains $p_+ \times \text{average positive P\&L}$ minus $p_- \times \text{average loss magnitude}$, using the actual positive and negative round frequencies. A favorable per-position payoff ratio does not guarantee favorable realized averages if losing rounds carry larger positions.

Final specification. Use target-payoff and target-average equations to derive order quantities; derive price ceilings from the same inventory and payoff state; let DIFF and the holdings ratio emerge from the desired outcome advantage; model joint price paths and execution; stage fills when necessary; and recalculate from confirmed state after every material change. Do not use a preset per-round capital limit as the sizing input.

The current screenshot supports the objective $C < \min(U, D)$, which produces positive conditional settlement in either direction. It does not establish that every round can reach that condition. When the reachable frontier cannot satisfy both-outcome profit, the controller must evaluate the actual remaining loss/profit trade-off rather than assume that additional capital alone guarantees recovery.

Sources

Supplied image: [451b1fe8-1621-4151-b6f1-360469d501c7.png](#). Dashboard values and plotted inventory, inspected directly. Round resolution is marked pending; order labels and averages are rounded.

1. Polymarket. Chainlink TWAP Prices. Accessed September 10, 2026. Reference feed windows and timestamp interpretation.
2. Polymarket. Fees. Accessed September 10, 2026. Published fee formula, market parameters and maker treatment.
3. Polymarket. Place Orders and Order Lifecycle. Accessed September 10, 2026. Post-only, tick/size requirements and order execution behavior.

Prior private specification: *Pasted markdown(2).md*, Wallet3048 v5. It supplies strategy descriptions and aggregate replay results, not the full event-level data needed for a market-optimal controller. All new target-price scenarios in this report are explicitly hypothetical.